

learned from? It is helpful to follow Graham and Dodd's path and trace the forces causing the discrepancy between price and value.

Perhaps the most significant of these factors over time has proven to be indexation. According to Bloomberg Intelligence,² the market share of passive equity strategies overtook that of active strategies around August 2018 and was nearly 55% by 2021, driven largely by the growth of funds tracking the S&P 500 and other broad U.S. indexes. This shift has been accelerated by a variety of forces, including the advent of robo-advisers, the rising adoption of model portfolios by advisers, and the growth in advisers' use of passive ETFs and funds. Active managers still dominate in fixed income, although passive's share rose to 34% from 31% in the third quarter of 2021 following the worst bond returns in decades amid rising rates and capital outflows. Given the significant amount of "closet indexing" in fixed-income markets, those percentages likely understate the true impact of index inclusion or exclusion on the universe of likely buyers (or forced sellers) for a given bond. When demand for a security is created by its mere inclusion on a list, without regard to its fundamental value, the price will change in ways that are not related to fundamental developments.

This brings us to the rating agencies—the organizations to whom passive market participants have outsourced the work that Graham and Dodd implore the assiduous analyst to undertake. The agencies have proven time and again to be backward looking, belatedly downgrading in response to past performance. Their ratings also demonstrate positive serial correlation, so we know they do not move their ratings enough each time to get to a place where the new rating reflects all available information at that moment. This reality makes it clear that spreads should and do move much faster than ratings and invite

² Bloomberg Intelligence, "Passive Likely Overtakes Active by 2026, Earlier if Bear Market" March 11, 2021, <https://www.bloomberg.com/professional/blog/passive-likely-overtakes-active-by-2026-earlier-if-bear-market/>.